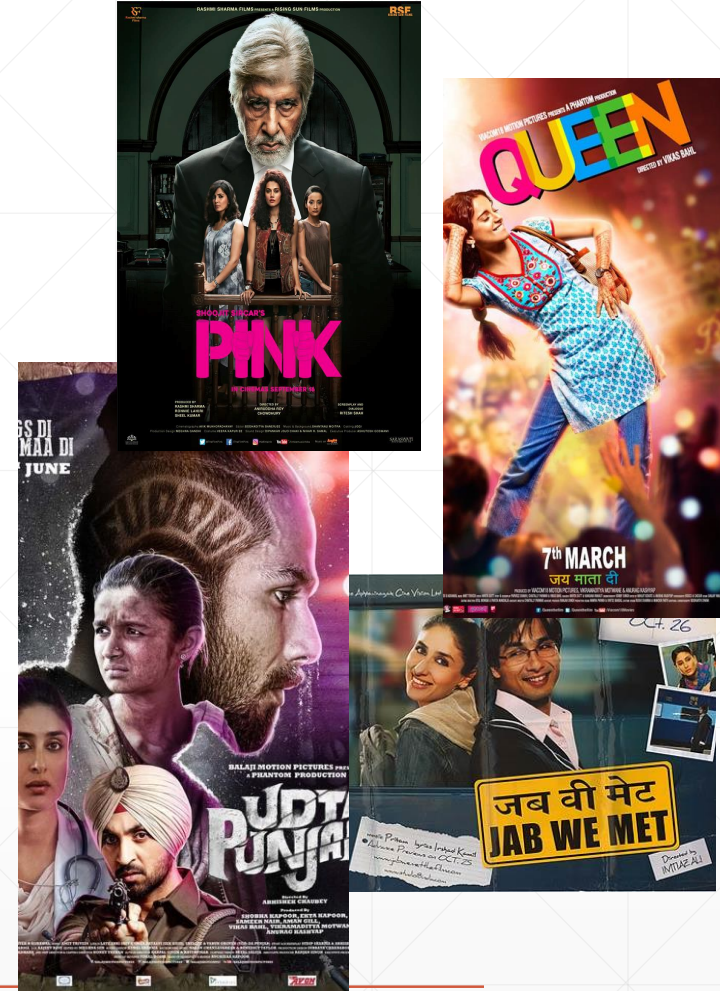




Quantitative Characterization of Code Switching Patterns in Complex Multi-Party Conversations: *A Case Study on Hindi Movie Scripts*

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An Example



ek minute ke liye thoda practical socho
Main tumharey angle se hi soch raha
hoon... Tum hi uncomfortable feel
karogi... bahut time ho gaya hai... bahut
fark aa gaya hai



Kismein? Mujhmein koi change nahin
hai



Vohi to baat hai... mujhmein hai... meri
duniya ... bilkul alag hai... ab... you'll not
fit in



Matlab? ek dum se main tumharey jitni
fancy nahin hoon...

Why are
scriptwriters
using code mixing?

What are the
patterns of code
mixing?

Prior Work

- Li Wei. 2002. *what do you want me to say? on the conversation analysis approach to bilingual interaction*.
 - Carol Myers-Scotton. 2005. *Multiple voices: An introduction to bilingualism*.
 - Yamuna Kachru. 2006. *Mixers lyricizing in hinglish: Blending and fusion in indian pop culture*.
 - Eva Losch. 2007. *The construction of social distance " through code-switching: an exemplary analysis for popular indian cinema*.
 - VINITI Vaish. 2011. *Terrorism, nationalism and westernization: Code switching and identity in bollywood*.
 - Subhash Chandra, Bhupendra Kumar, Vivek Kumar, and Sakshi. 2016. *Acute sporadic english in bollywood film songs lyrics: A textual evidence based analysis of code-mixing in hindi*.
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Takeaways from Linguistic Studies

Code mixing for,

- Specific communicative purpose
 - Social identity
 - Linguistic style accommodation
 - Encoding social distance
 - Borrowing fresh metaphors and new rhyming words from English
 - Westernization of culture
-

Our contributions

- Developed a set of **quantitative** conversation analysis techniques for code switching
 - A few **visualization techniques** for code switching to get a holistic view of switching styles in interactions
 - First quantitative analysis of code switching patterns in **Hindi movies**
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Outline



Outline



Data

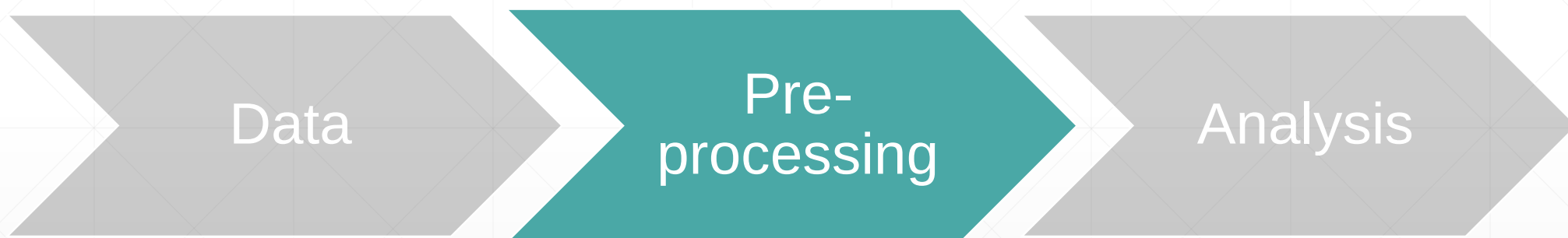
Source:

<https://moifightclub.com/category/scripts/>

ID	Movie	Year	# Turns
1	Ankhon Dekhi	2014	753
2	D-Day	2013	659
3	Dedh Ishqiya	2014	642
4	Dum Laga Ke Haisha	2015	678
5	Ek Main Aur Ekk Tu	2012	836
6	Kapoor and Sons	2016	1119
7	Kai Po Che	2013	675
8	Lootera	2013	734
9	Masaan	2015	653

ID	Movie	Year	# Turns
10	Neerja	2016	602
11	NH10	2015	340
12	Pink	2016	897
13	Queen	2014	951
14	Raman Raghavan 2.0	2016	373
15	Shahid	2013	896
16	Talvar	2015	823
17	Titli	2015	656
18	Udaan	2010	955

Outline



Pre-processing

- Separating characters, dialogues and scenes
- Language labeling (Gella et al., 2013)
- Errors
 - PDF justification issues
 - Identifying the characters
 - Language tagger

EXT. A WEDDING PANDAL - DAY

1

PREM, 24, a lanky young man - dressed in his school uniform is sitting across the wedding pyre. A girl is sitting next to him with her veil down till her shoulder - her face is not seen. Prem's family sits behind him peeping eagerly towards the wedding processions going on. In the background, latest hit of the 90s plays at a faster rpm.

THE PRIEST
(in strange English)
You are in a wedding hoga, ya You
are at a wedding hoga?

PREM
In - in a wedding.

THE PRIEST
Galat jajman - at a wedding.

Prem's father slaps him from behind.

PREM'S FATHER
Ek sawaal na theek jawaab diya
jaaye tere se!

THE PRIEST
Main bina apni biwi ke kabhi upwan
na jaaunga. English mein translate
karien?

PREM
I...I not...

PREM'S FATHER

Outline



Projections

- $\mathcal{C}$: entire corpus, $\mathcal{P}$: all participants in $\mathcal{C}$
 - **Corpus level**
 - $\pi(\mathcal{C})$: set of utterances in the entire corpus
 - **Participant level**
 - $\pi_{P_i}(\mathcal{C})$: set of all utterances of the participant $P_i \in \mathcal{P}$
 - **Dyad level**
 - $\pi_{D_{i,j}}(\mathcal{C})$: set of all utterances of the dyad $(P_i, P_j), P_i, P_j \in \mathcal{P}$
 - dyad is two consecutive utterances of P_i, P_j but not necessarily in that order
-

Metrics

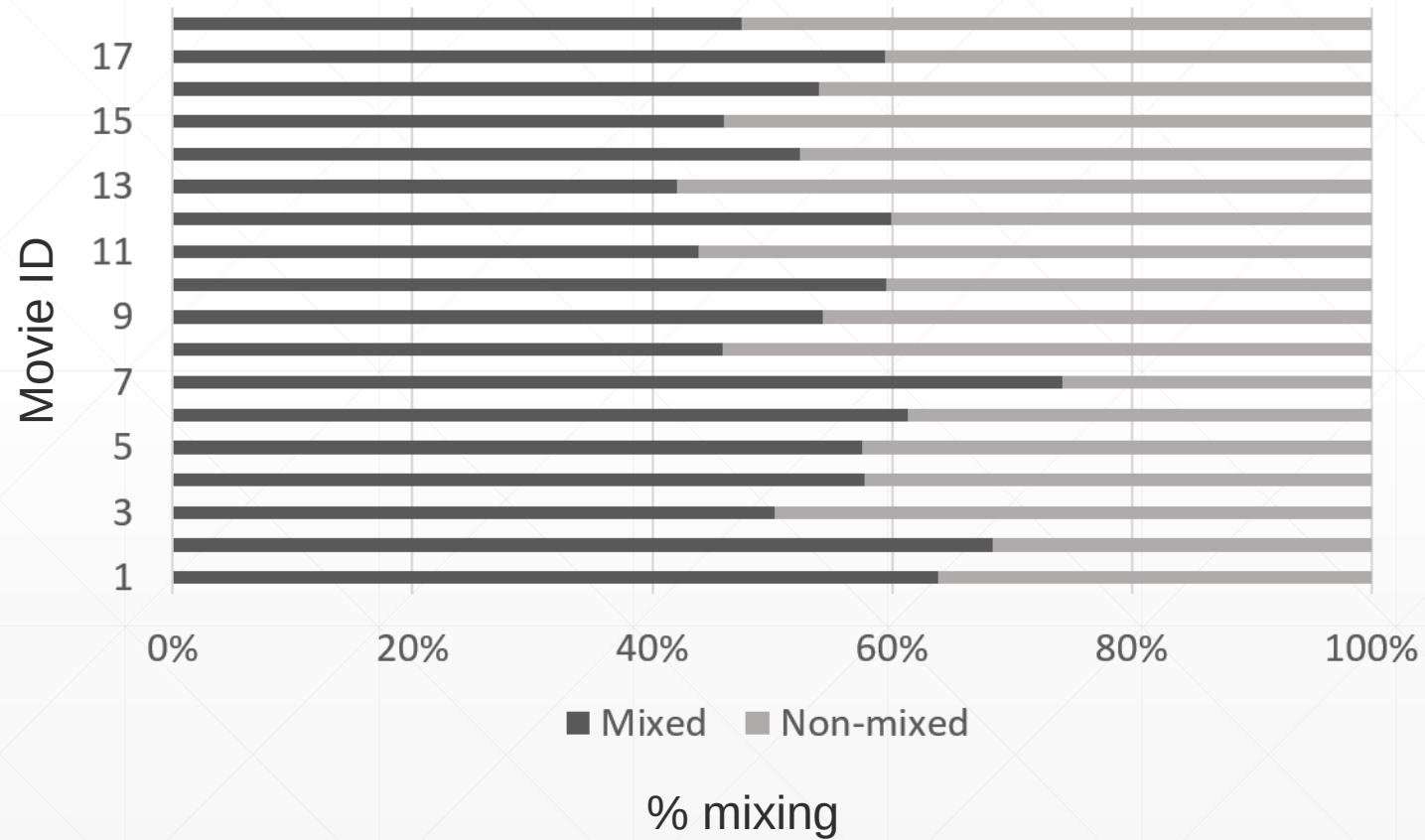
Gamback and Das, 2016

- Code Mixing Index

Guzman et al., 2017

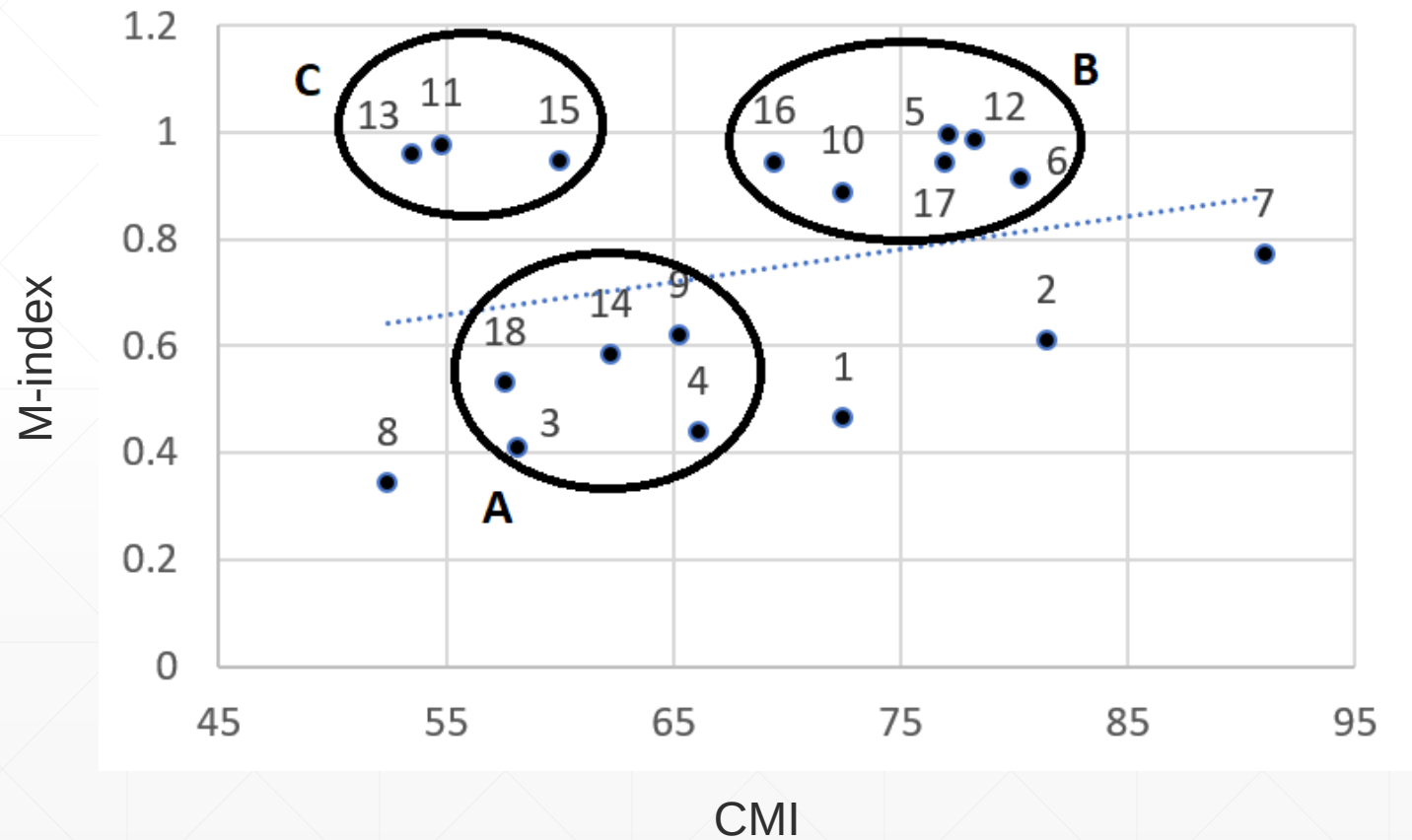
- Multilingual (M) Index
 - Language Entropy
 - Probability of Switching
 - Burstiness
 - Span Entropy
 - Memory
-

Code Mixing Aggregates



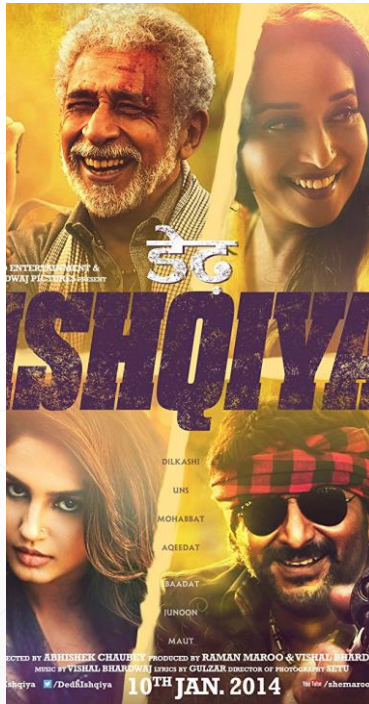
Socio-cultural setting of the film

Corpus Level



- **A**
 - Low CMI and M-index
- **B**
 - High CMI and M-index
- **C**
 - Low CMI but high M-index

Corpus Level

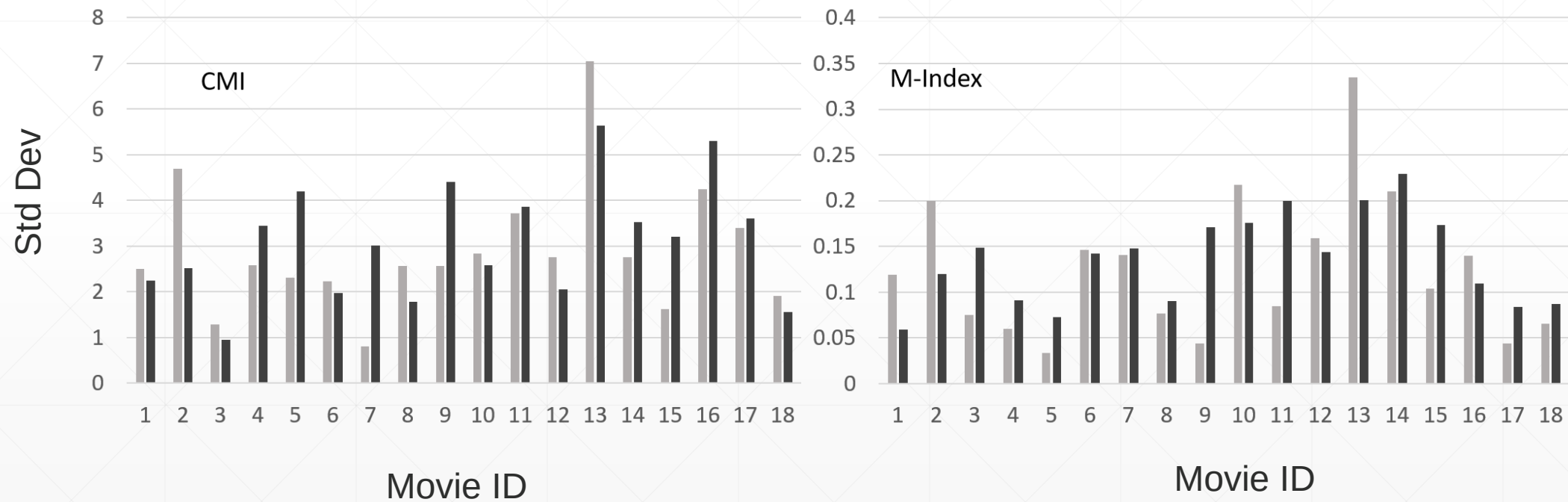


A: small towns, period

C: multiple matrix languages

B: urban

Participant Level



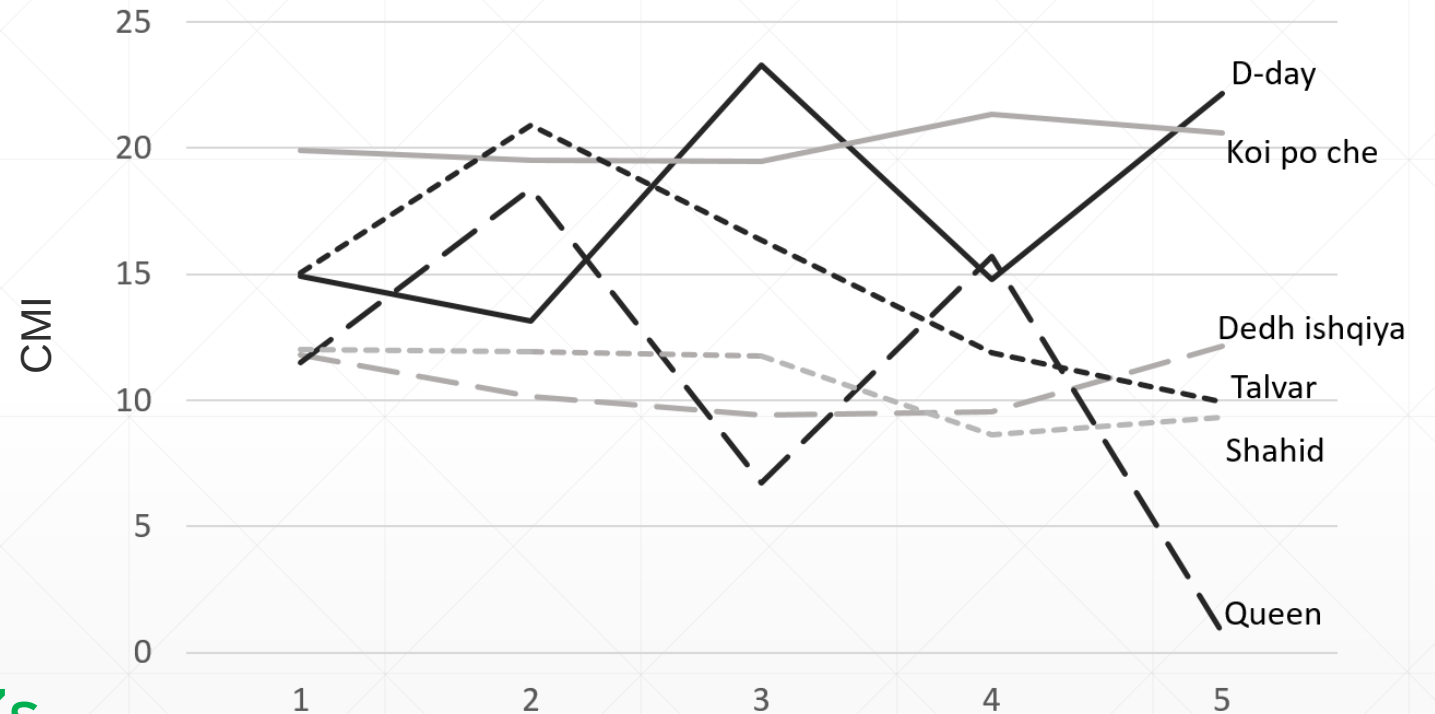
light grey: character, dark grey: dyad

Participant Level

Queen

- Rani, Vijay and Mom
- Vijaylaxmi and Mikhaelo

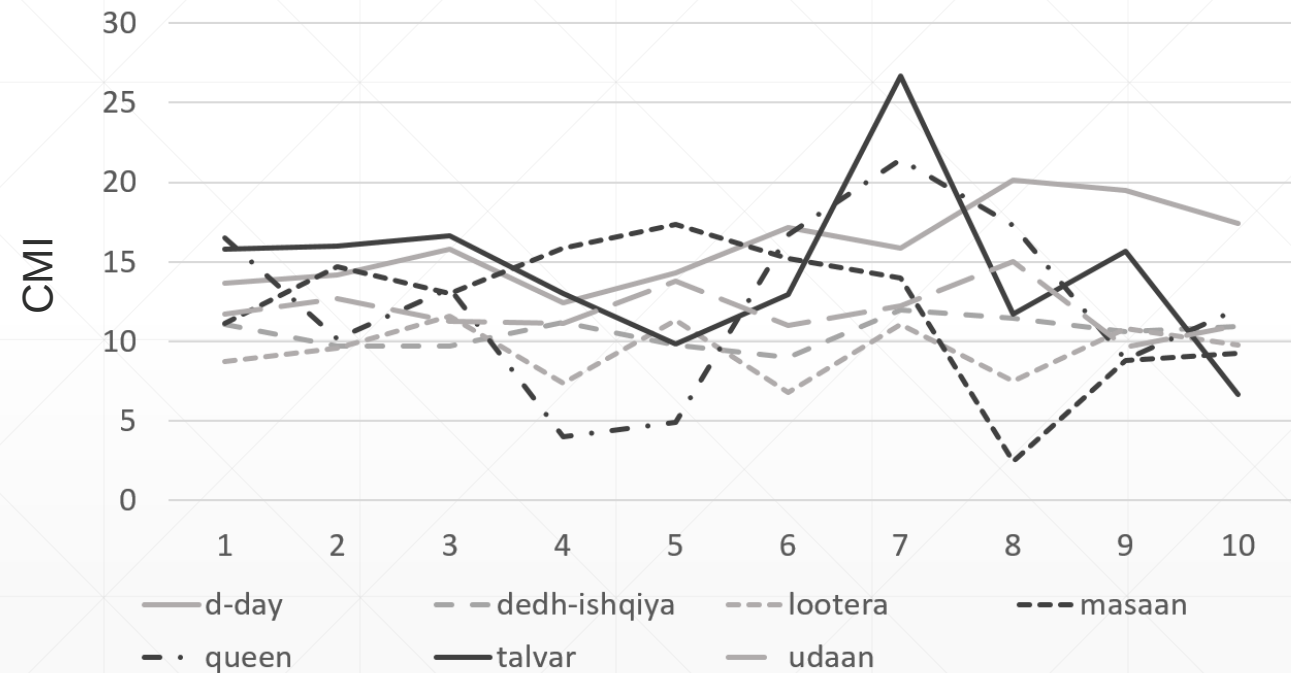
Establishing the participant's identity



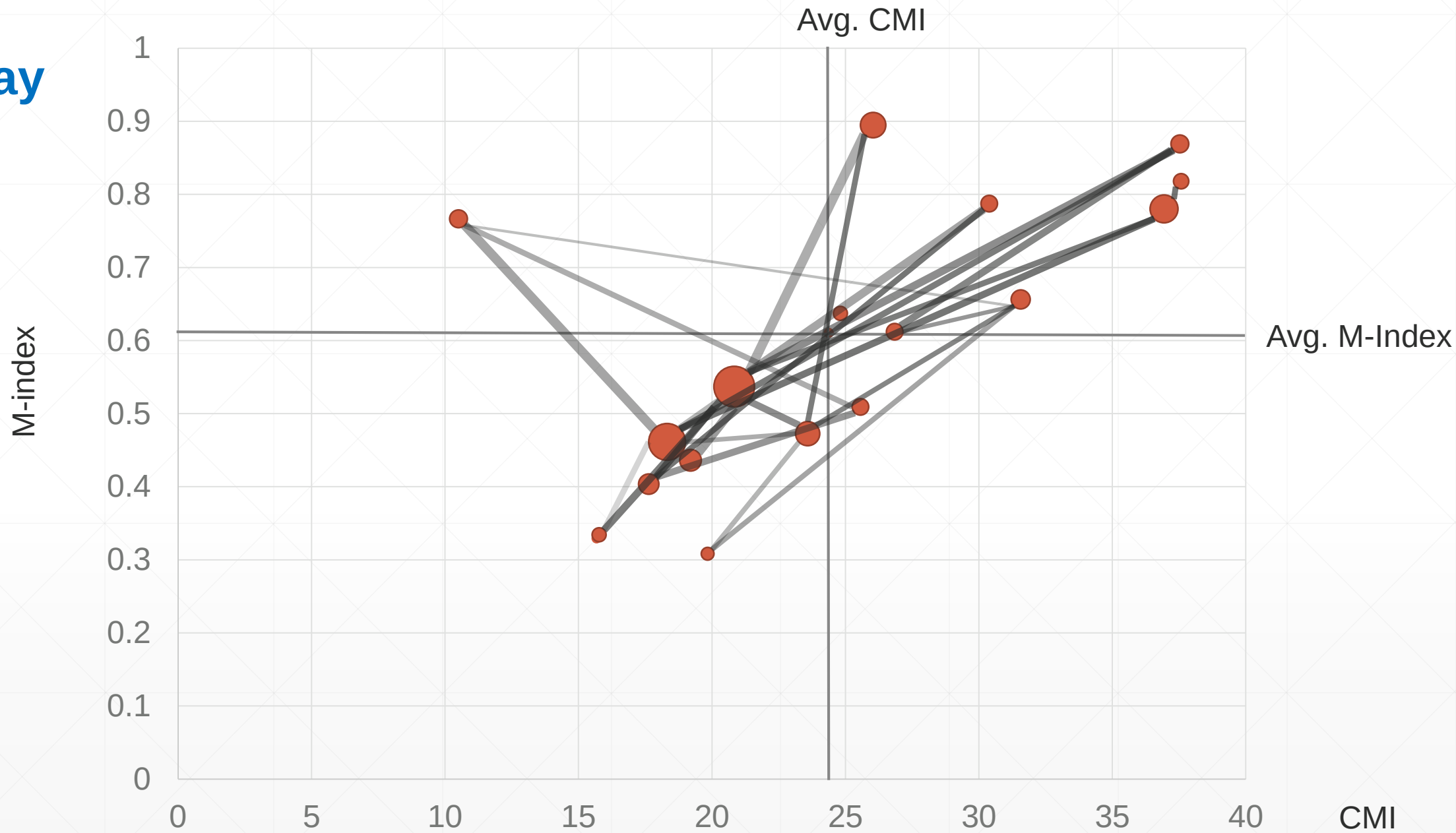
Dyad Level

- Queen
 - High variance in characters
 - High variance in dyads
- D-day
 - High variance in characters
 - Low variance in dyads

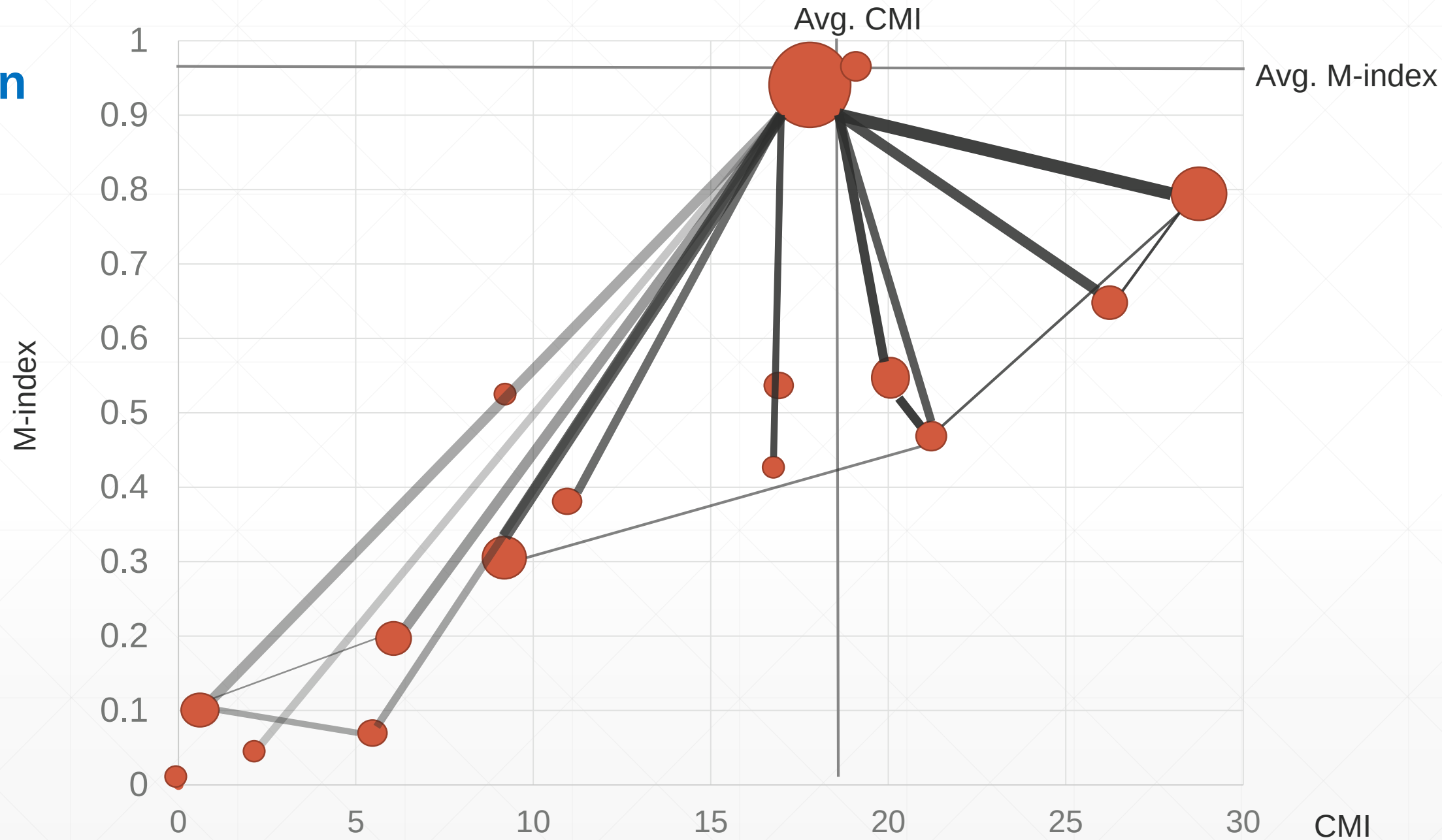
Participant's social graph



D-day



Queen

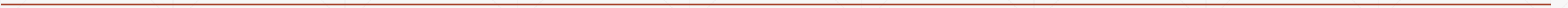


Conclusion

- A framework for quantitative characterization of CS patterns in multi-party conversations, beyond the corpus level footprints
 - Illustrates the use of code-mixing to establish
 - Social setting of the film
 - Character's identity
 - Relations among characters
-

Future Work

- Applying our framework to other conversational data
- Studying linguistic style accommodation
- Observing pragmatic and structural aspects of CS



Thank you

Questions?

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